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AUTOMATIC PAIN RECOGNITION SYSTEM FORDENTALPATIENTS USING MACHINE LEARNING

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1 Abstract

Dental patients often struggle to effectively communicate their pain, especially during procedures, relying on gestures and physical movements. This may disturb the treatment process by the dentists. Current method relies heavily on patient self-reporting, which can be subjective and inconsistent. In this study, we present a machine learning-based automatic pain recognition system designed to objectively recognise the pain levels in dental patients. Machine learning models were trained in Python using Jupyter Notebook with various algorithms like Random Forest and Logistic Regression. The system analyzed bio-signals like ECG, EMG, and EEG via sensors, categorizing them into "High pain," "Mild pain," and "No pain." Utilizing eight algorithms, the Random Forest model achieved the highest accuracy of 65.076%. While data collection presented challenges and further clinical validation is necessary, this research presents a promising step towards intelligent healthcare systems that could improve pain management and treatment efficacy in dentistry.

2 Research Problem & Objectives

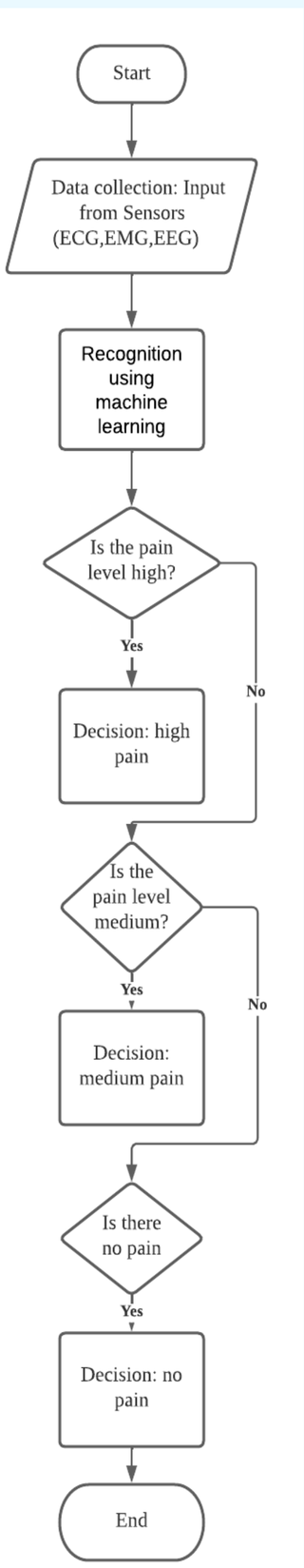
Problem Statement:

Dentists often encounter challenges when patients struggle to effectively communicate dental pain during surgery or examinations. To address this issue, a machine learning-based automatic pain recognition system is needed. Such a system can identify and assess a patient's pain level.

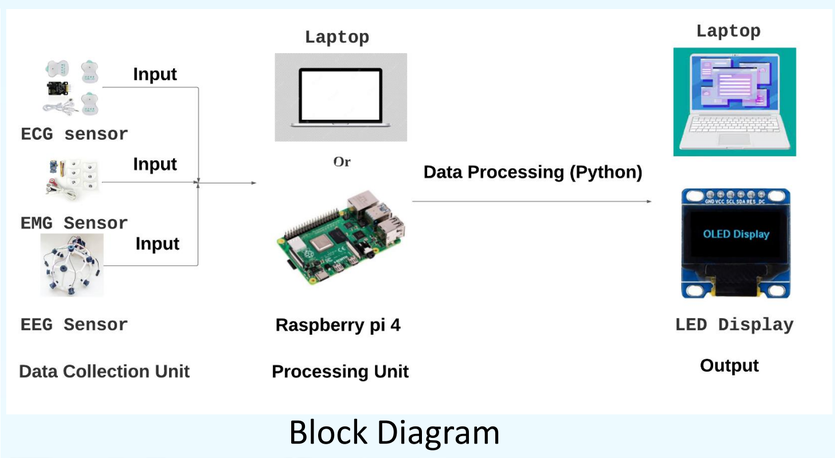
Objectives:

* To develop a system for collecting bio-signal data for dental pain detection during treatment.
* To develop a machine learning based system to automatically recognize patient's pain level.
* To test the developed automatic pain detection.

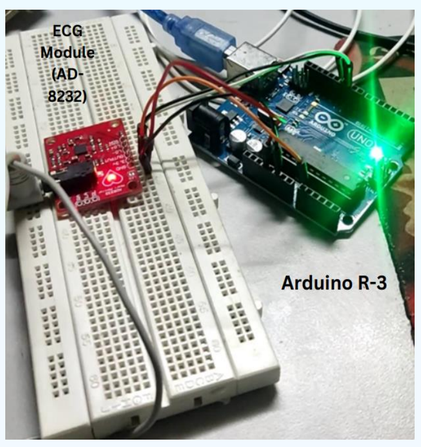
3 Methodology



Flow Chart

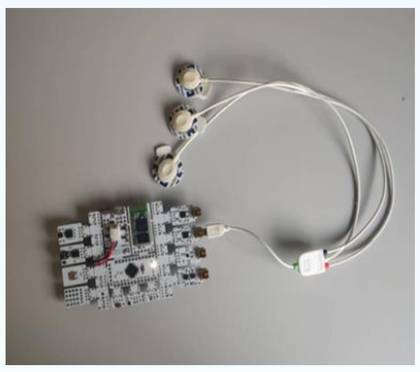


Block Diagram

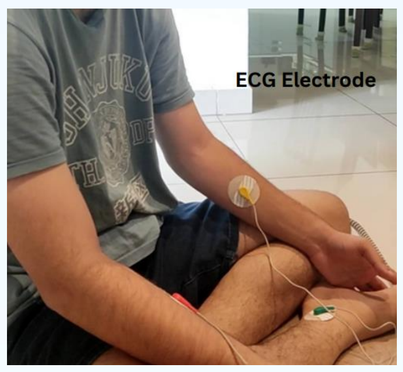


ECG Sensor Setup

EEG Sensor placement



BITalino Controller(EMG)&(EEG)



ECG Sensor placement

* ECG sensors on hands monitor heart activity.
* EMG sensors on jaw muscles gauge muscle activity during discomfort.
* EEG sensors on the scalp measure brain waves related to pain perception.
* Six sets of ECG, EMG, and EEG data gathered meticulously.
* 80% of data used for training, enabling pattern recognition.
* 20% reserved for testing model accuracy post-training.
* Rigorous testing with random data to ensure generalization and prediction accuracy beyond the training dataset.

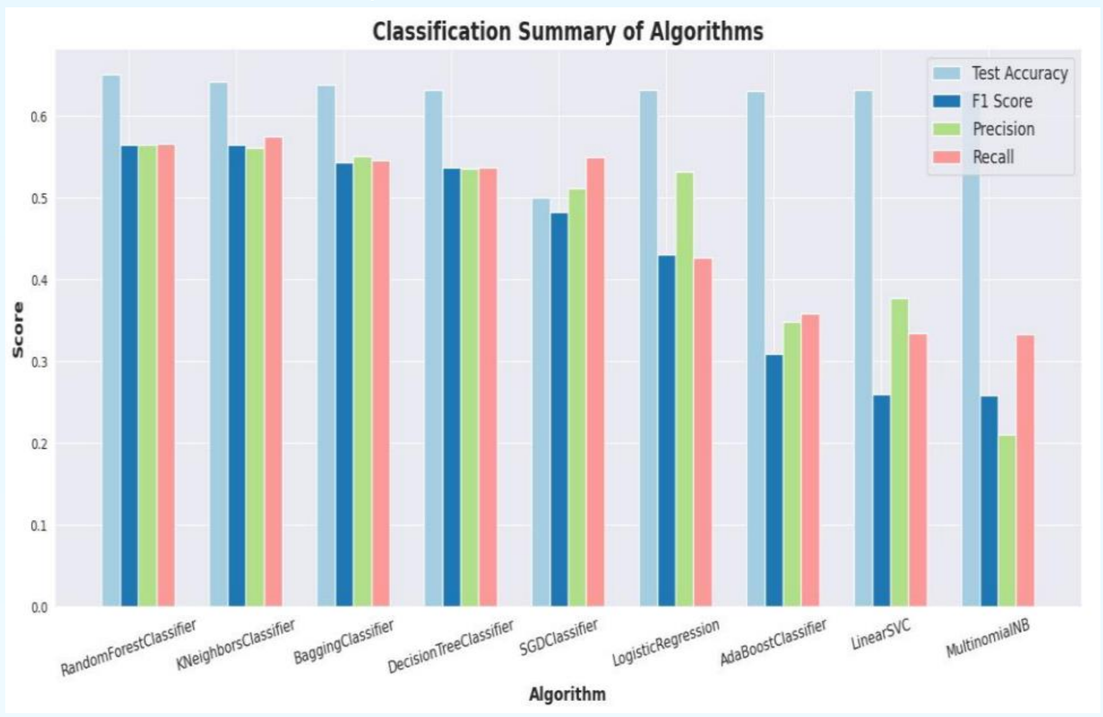
4 Results & Discussions

	Algorithm	Accuracy: Test	Precision: Test	Recall: Test	F1 Score: Test
0	RandomForestClassifier	0.65052	0.564671	0.565391	0.564855
1	KNeighborsClassifier	0.64200	0.561086	0.574755	0.564362
2	BaggingClassifier	0.63760	0.550880	0.545020	0.542867
3	DecisionTreeClassifier	0.63132	0.535953	0.536837	0.536392
4	SGDClassifier	0.50052	0.511900	0.549750	0.481757
5	LogisticRegression	0.63216	0.531343	0.426353	0.430173
6	AdaBoostClassifier	0.63004	0.348132	0.357687	0.308544
7	LinearSVC	0.63120	0.377098	0.334044	0.259729
8	MultinomialNB	0.63120	0.210400	0.333333	0.257970

Model Accuracy

Analog ECG Signal

Analog EEG Signal



* Pain level recognition involves using six sets of input data from four different sensors for training and testing purposes.
* Each set comprises examples derived from data collected from multiple individuals who are categorized as experiencing no pain, mild pain, or high pain.
* The input values consist of heart rate data (ECG), muscle activity data (EMG), and four brain signal data (EEG) recorded from each person.
* The output class indicates the perceived level of pain for each individual.
* The input data from the sensors are utilized to train a model that can recognize and classify pain levels based on the recorded physiological signals.
* The model aims to learn patterns and correlations between the physiological data and the corresponding pain levels to accurately predict pain perception.
* Random forest: 65.076% accuracy, prediction time of 2.117676s.
* Decent recall: 0.562166, indicating dataset sensitivity.
* Impressive F1 score: 0.563741, balancing precision and recall.

5 Conclusions

* System achieved a partial success with 65.076% accuracy in predicting pain levels.
* Incorporating medical-grade equipment promises enhanced accuracy. Testing with actual patient data can boost accuracy and effectiveness.
* Using specialized medical tools can significantly improve reliability for real patient outcomes.
* Enhanced accuracy in clinical settings will amplify the system's practicality and overall effectiveness.

Future Work:

*Replace simulated dental patient data obtained from real patients in a hospital environment for heightened result accuracy.
* Medical-grade equipment for reliable data collection.
* Integrate diverse sensors for comprehensive patient-friendly data collection.